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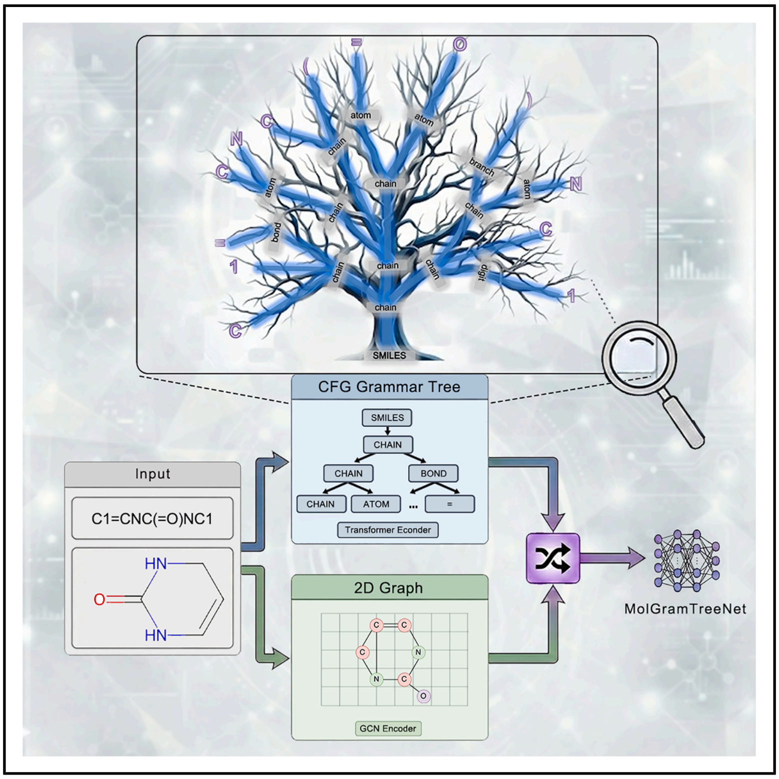
Graphical abstract



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In brief

Molecules; Computational chemistry; Materials chemistry

Highlights

- MolGramTreeNet fuses grammar trees and graphs for molecular property prediction
- A dual-path framework captures chemical rules and hierarchical structures explicitly
- The model achieves superior performance across ten molecular property benchmarks
- Attention visualization provides interpretable insights into toxic functional groups

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MolGramTreeNet: A multimodal molecular property prediction model via grammar tree-constrained molecular representation

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SUMMARY

Molecular property prediction is pivotal for drug discovery. However, existing methods that rely on linear SMILES or 2D graphs often overlook explicit hierarchical structures and chemical grammar constraints. To address this limitation, we propose MolGramTreeNet, a multimodal framework that leverages grammar tree-guided representations. The framework parses SMILES into hierarchical trees using context-free grammar, explicitly encoding chemical rules. The architecture employs a dual-path encoder: a Transformer captures tree-based hierarchical semantics, while a Graph Convolutional Network extracts 2D topological features. These fused features generate comprehensive molecular representations. Experiments conducted across ten datasets demonstrate that MolGramTreeNet outperforms eight baseline models, achieving performance improvements of 2.86%–21.47%. Ablation studies confirm the grammar tree's contribution. This approach significantly enhances both accuracy and interpretability, thereby advancing AI-driven drug discovery.

INTRODUCTION

Molecular property prediction is a core task in computational chemistry and drug discovery, with the goal of efficiently predicting key properties such as solubility, toxicity, and biological activity through molecular structures.[1] Accurate prediction of molecular properties facilitates the identification of antiviral candidates and enhances clinical trial success rates, ultimately improving therapeutic outcomes. Such studies not only facilitate the discovery of novel therapeutics and the repurposing of existing drugs but also contribute to optimizing the use of existing drugs, providing additional options and opportunities for combating viruses. contribute to optimizing the use of existing drugs, providing additional options and opportunities for combating viruses.

With the rapid advancement of deep learning technology, molecular representation learning has emerged as one of the core challenges in this field. The choice of molecular descriptors significantly influences the performance of deep learning models. Drawing inspiration from natural language processing, a widely adopted molecular sequence representation method employs a string format known as the Simplified Molecular-Input Line-Entry System (SMILES) to describe molecular structures. SMILES serves as a linear notation for inputting and representing molecular structures. By establishing an analogy between SMILES and text, and leveraging the success of sequence models in natural language processing, sequences can similarly be processed as textual input.[2,3] Building upon sequence models with SMILES input, a language model for SMILES expressions can be constructed using recurrent neural networks (RNNs) to automatically generate meaningful SMILES representations. To capture long-range dependencies, architectures such as Bidirectional Long Short-Term Memory (BiLSTM) networks have been widely adopted for extracting drug SMILES.[4,5] Beyond directly inputting SMILES sequences into RNN-based models, the word2vec model in natural language processing treats each symbol in molecular expressions as a word, enabling the learning of molecular substructure embeddings from SMILES data.[6,7] Additionally, molecular representations can be derived by training sequence-to-sequence (Seq2Seq) models using various molecular descriptors as input.[8,9] Unsupervised pre-training of sequence language models has facilitated the development of architectures such as SMILES Transformer, which can learn molecular fingerprints for downstream tasks.[10] Following the emergence of BERT in natural language processing, the research paradigm has increasingly shifted toward fine-tuning pre-trained models.[11–14] However, purely text-based SMILES representations exhibit notable limitations: their linear encoding fails to explicitly capture molecular hierarchical structures, such as branched and cyclic substructures. This requires the model to autonomously learn complex SMILES grammatical rules to generate valid molecules, which not only increases training difficulty but may also yield chemically implausible predictions due to insufficient grammatical constraints.

Due to the limitations of SMILES linear encoding in representing the intrinsic structure of molecules, graph neural network (GNN) molecular graph models have been extensively studied. These models represent molecules as graphs, where atoms or functional groups serve as nodes and chemical bonds as edges.[15–20] Subsequently, graph convolutional networks (GCN) are used to encode molecular graphs based on atomic features.[21] Attention mechanisms have been integrated into graph neural networks to enhance molecular representation for drug discovery.[16,22–24] The message passing neural network (MPNN) model performs an edge-based message-passing process on directed graphs, generating vector representations of atoms and bonds.[25] The scarcity of labeled data remains a major challenge for GNN models in predicting molecular properties. Graph contrastive learning (GCL) methods have demonstrated strong performance in applications with limited labeled data.[26,27] However, traditional GNNs rely on adjacency matrices that focus on topology but lack explicit constraints based on chemical grammar rules, such as valence validity and functional group connectivity. Additionally, they struggle to directly encode the hierarchical structural information embedded in SMILES, such as the recursive generation rules of cyclic structures. Furthermore, models relying solely on sequence features or graph structure features face dimensional limitations in semantic representation. The linear sequence of SMILES struggles to parse nested molecular substructures, while the adjacency representation of two-dimensional graphs overlooks the hierarchical combination logic of chemical functional groups.

Beyond these foundational GNN models, the field has witnessed significant advancements aimed at enhancing molecular representation by incorporating richer information. Some approaches focus on integrating 3D spatial and geometric information, which is lost in 2D graphs; for instance, AEGNN-M utilizes a 3D graph-spatial co-representation model to leverage this high-dimensional data.[28] Other methods aim to refine the 2D graph representation itself, such as MS-BACL, which introduces bond graph augmentation and contrastive learning,[29] or CardiOT, which explores novel architectures such as optimal transport (OT) and Kolmogorov-Arnold networks (KANs).[30] Simultaneously, several multimodal approaches that fuse sequence (1D) and graph (2D) information have demonstrated strong performance. For example, GraSeq integrates graph features from a GCN with sequence features from a Bi-LSTM to achieve improved prediction.[31] Other studies have explored similar fusions, such as combining LSTM and GAT,[32] or learning joint representations from both SMILES and graphs.[33] Concurrently, hybrid models such as TranGRU have focused on capturing both local and global molecular information.[34] While these methods—whether enhancing 3D/bond graphs or fusing 1D/2D data—effectively combine richer data, they inherit the limitation of failing to explicitly enforce chemical grammar rules or capture the inherent hierarchical nature of molecules, a key limitation shared with earlier sequence-only models.

To address these limitations, we propose MolGramTreeNet, a multimodal framework that utilizes grammar tree-guided representations for molecular property prediction. By explicitly incorporating Context-Free Grammar (CFG) parsing technology, the SMILES sequence is transformed into a hierarchical grammar tree representation, thereby providing richer chemical semantic information and rigorous grammar constraints for molecular property prediction. From a theoretical perspective, Vipul et al. demonstrated, through information theory, that grammar tree representation enhances model accuracy and reduces parameter count by minimizing conditional entropy.[35] In subsequent work, they introduced a grammar ontology-driven Transformer framework, successfully integrating molecular grammar rules and hierarchical structure representation in chemical reaction outcome prediction, which provided significant inspiration for this study.[36] Research by Kusner et al. within the Bayesian framework further validated the efficacy of this approach, achieving the optimization of single-molecule properties for drug-like molecules through grammar-based representation.[37] Additionally, the Supervised Grammar VAE proposed by Oliveira et al. effectively combined grammar tree structure with deep generative models, demonstrating notable advantages in molecular property prediction and controllable molecular design tasks.[38] The G-MATT model further expanded the application scope of grammar trees.[39] By constructing a tree-sequence converter architecture based on SMILES grammar trees, this approach achieved significant improvements in the accuracy of single-step retrosynthesis prediction. The core innovation of these models lies in their utilization of molecular grammar trees as an intermediate representation to bridge chemical knowledge and data-driven models. This methodology not only prevents the generation of invalid structures but also captures the spatial dependencies of substructures through tree-like hierarchies while retaining a manageable parameter count of Transformer architectures, further demonstrating the potential of grammar tree representations in molecular intelligent modeling. Specifically, the SMILES string is first parsed into a grammar tree via context-free grammar (CFG), explicitly decomposing it into hierarchically meaningful chemical units such as functional groups, cyclic structures, and reaction sites. For instance, "CC(=O)O" can be parsed into a tree structure with the carboxylic acid group (=O and O) as the core and the alkyl chain (C-C) as the branch. This representation retains sequential characteristics while explicitly capturing the spatial dependencies of substructures through hierarchical tree structures, thereby fundamentally preventing the generation of invalid molecular structures. To date, based on a synthesis of prior research, molecular grammar tree representations have been applied to downstream applications such as molecular optimization, single-step retrosynthesis prediction, and molecular property prediction using the QM9

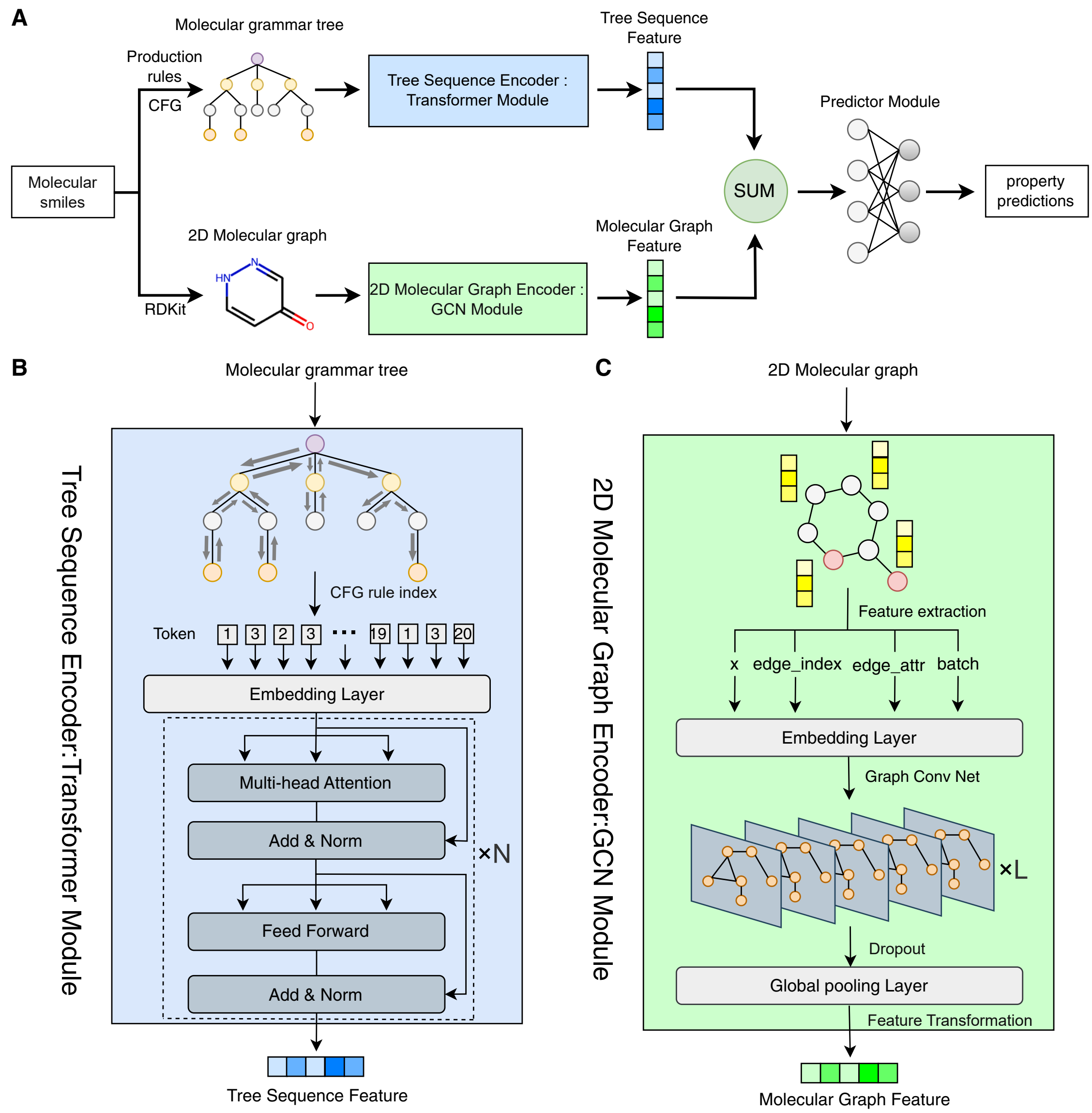

Figure 1. The framework of MolGramTreeNet

(A) Overview of the overall workflow of MolGramTreeNet.

(B) Schematic diagram of Tree Sequence Encoder: Transformer module.

(C) Schematic diagram of 2D molecular graph Encoder: GCN module.

dataset. However, for other datasets in MoleculeNet, no suitable model has yet been identified to perform molecular property prediction tasks using grammar tree representations. Therefore, this study aims to design a molecular property prediction model based on a grammar tree representation.

To achieve complementary fusion of multidimensional features, MolGramTreeNet employs a dual-path encoding mechanism. Rather than relying on a single modality, the architecture integrates hierarchical semantic features extracted from grammar trees (via a transformer) with spatial topological features derived from molecular graphs (via a GCN). This multimodal strategy explicitly incorporates chemical grammar rules into representation learning, generating comprehensive molecular embeddings that enforce chemical validity while capturing complex structural dependencies. By bridging the gap between sequential grammar and graph topology, this approach significantly enhances the chemical rationality of the prediction model compared to traditional single-view methods.

The core contributions of this study are as follows.

1. Explicit Grammar Encoding: We integrate CFG-based grammar trees to explicitly encode hierarchical structures and chemical rules, ensuring chemically valid representations.
2. Development of a Multimodal Feature Fusion Framework: We constructed a dual-path framework that integrates

Table 1. Simplified molecular SMILES syntax

No.	Production rules
1	SMILES → chain
2	chain → chain branched_atom
3	chain → chain bond branched_atom
4	chain → branched_atom
5	branched_atom → atom ringbond
6	branched_atom → atom
7	branched_atom → atom BB
8	branched_atom → atom RB
9	BB → branch
10	RB → ringbond
11	branch → (chain)
12	ringbond → digit
13	bond → =
14	atom → aromatic_organic
15	atom → aliphatic_organic
16	aliphatic_organic → C
17	aliphatic_organic → O

Transformer-encoded SMILES sequences and GCN-encoded molecular graphs. By employing channel concatenation, we achieved cross-dimensional interaction to generate a comprehensive representation fusing chemical semantics, spatial topology, and grammatical information.

3. Enhanced Prediction and Interpretability via Feature Compression: We established a robust feature-prediction correlation by compressing multidimensional features into low-dimensional vectors. This approach significantly improves accuracy across ten benchmarks and provides interpretable insights for drug design, explicitly identifying toxicophores through attention visualization.

By integrating multidimensional information from sequences, graph structures, and grammar trees, MolGramTreeNet introduces a novel paradigm for molecular property prediction that combines chemical rationality with computational efficiency. This advancement is expected to facilitate the progression of data-driven drug discovery toward greater accuracy and interpretability.

RESULTS

MolGramTreeNet

Figure 1 illustrates the MolGramTreeNet framework, which introduces a multimodal architecture that synergizes molecular grammar constraints with topological representations. As shown in the overall workflow in Figure 1A, the model processes molecular SMILES through two parallel pathways. The first pathway employs the simplified Context-Free Grammar (CFG) rules defined in Table 1 to parse SMILES strings, capturing their hierarchical chemical semantics. Figure 2 presents a representative case study of this process, demonstrating how the construction of a molecular grammar tree explicitly encodes nested substructures. Subsequently, the rule indexing mechanism illustrated in

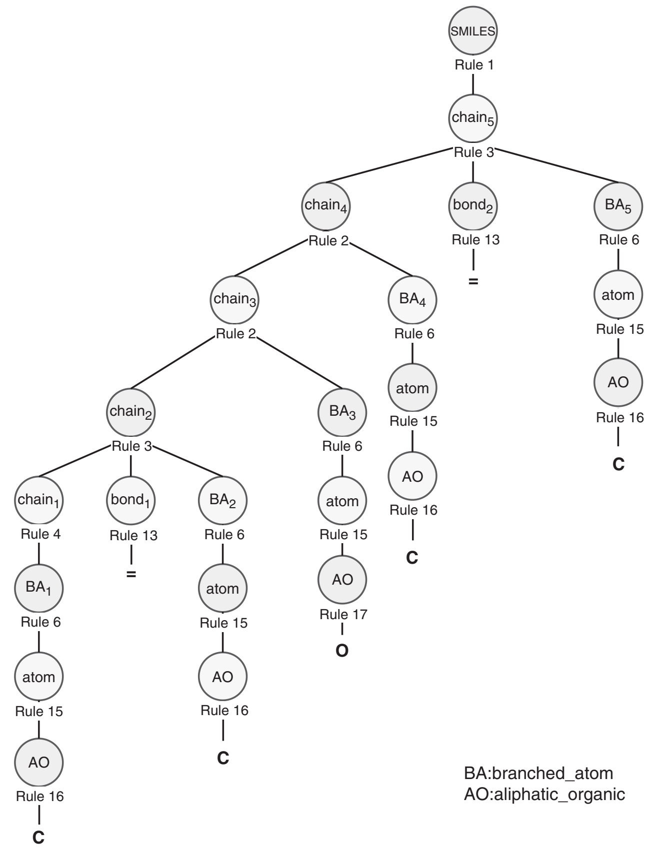

Figure 2. The parse tree of the simplified divinyl ether SMILES (C=COC=C) generated using the simplified molecular SMILES rule syntax presented in Table 1

The numeric subscripts (e.g., $chain_1$ and $branched\_atom_2$) associated with non-terminal nodes function as sequential and positional identifiers. These subscripts explicitly indicate the order of generation and the hierarchical relationships among nodes of the same type during the bottom-up construction process, thereby ensuring traceability and eliminating structural ambiguity.

Figure 3 flattens this tree structure into a discrete token sequence for deep learning. The Transformer-based module detailed in Figure 1B then encodes this sequence to extract grammatical features. Concurrently, the second pathway, shown in Figure 1C, transforms the SMILES into a 2D molecular graph using RDKit and extracts topological features via a graph convolutional network (GCN). Finally, these complementary grammatical and topological features are fused to enable robust molecular property prediction. Specific implementation details and algorithmic procedures are provided in the STAR Methods section.

To provide a clear context for our model's performance, we first quantify its computational requirements. Our MolGramTreeNet model consists of approximately 45.8 million trainable parameters. A detailed breakdown of the model architecture, hyperparameters, hardware used, and measured training throughput (e.g., training time per epoch) is presented in the Supplementary Information, Table S2. This quantification demonstrates that our model achieves its results with a practical and competitive computational budget.

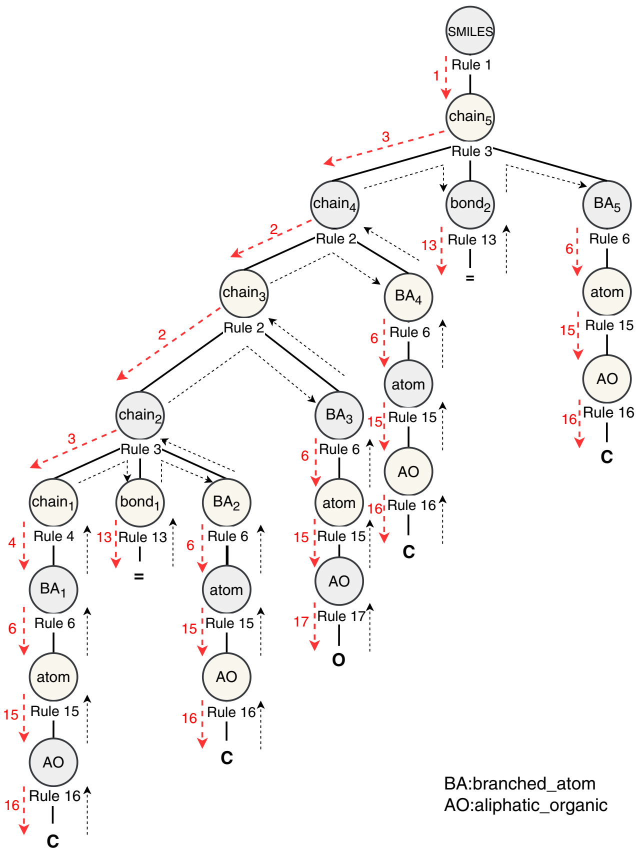

Figure 3. Rule index extraction from the grammar tree of "C=COC=C."

This figure illustrates the top-down, left-to-right traversal process for converting the hierarchical tree structure (Figure 2) into a sequence of production rule indices. The extraction proceeds by recursively applying and recording the rules from the root node down to the terminal nodes.

Overall performance of MolGramTreeNet on classification and regression benchmarks

Table 2 presents the calculated area under the receiver operating characteristic curve (ROC-AUC) for MolGramTreeNet on classification benchmarks, while Table 3 displays the root-mean-square error (RMSE) performance of the evaluated model and baseline methods on regression benchmarks. To provide a more comprehensive evaluation of model performance, we have also included eight additional evaluation metrics (e.g., F1-score, ACC, R^2, MAE, and so forth) for all ten datasets in the Supplementary Information, as detailed in Table S5. All models were re-evaluated under identical laboratory conditions. Datasets were randomly partitioned, with the best-performing model indicated in bold and the second-best result underlined. As shown in Table 2, MolGramTreeNet achieved the highest performance on three classification task datasets: ClinTox, Tox21, and Toxcast, and the second-highest performance (tied with KCL) on BBBP.

For regression tasks, Table 3 demonstrates that MolGramTreeNet attained the highest performance on ESOL and competitive performance on Lipophilicity, although several baseline models exhibited lower RMSE values on the latter dataset. This variation highlights the complementary strengths of different modeling approaches across diverse molecular properties. Specifically, MolGramTreeNet outperformed eight baseline models across ten tasks, with average performance improvements ranging from 2.86% to 21.47%. On the ESOL dataset, performance improved by 18.36% relative to the suboptimal model KCL, significantly enhancing screening capability for water-soluble molecules. On the ClinTox dataset, the ROC-AUC reached 0.994, surpassing KANO (0.926) and KCL (0.919). Compared with the suboptimal model KANO, performance improved by 7.34%, substantially enhancing the ability to identify clinically toxic molecules. The detailed methodology for calculating these performance improvement percentages is provided in the supplemental information under "Performance Calculation Methodology."

3D-extended MolGramTreeNet for quantum property prediction

To evaluate the impact of MolGramTreeNet on molecular three-dimensional structural information, we conducted an extended study on the original model. The enhanced MolGramTreeNet+3D integrates a computationally efficient three-dimensional encoder capable of simultaneously processing atomic coordinates and elemental information. The specific details of this model are documented in the STAR Methods section.

We evaluated the performance of MolGramTreeNet+3D on five quantum mechanical properties from the QM9 dataset, where 3D structural information demonstrates particular relevance: dipole moment (μ), polarizability (α), HOMO energy (ϵ_{HOMO}), LUMO energy (ϵ_{LUMO}), and HOMO-LUMO gap ($\Delta\epsilon$).

Table 4 summarizes the comparative experimental results, with optimal performance indicated in bold and secondary results underlined. Our 3D-extended model achieves competitive performance on multiple electronic properties. Specifically, MolGramTreeNet+3D reduces MAE on ϵ_{HOMO} by 4.5% compared to KANO (0.0042 vs. 0.0044 eV), on ϵ_{LUMO} by 23.8% (0.0048 vs. 0.0063 eV), and on $\Delta\epsilon$ by 4.9% (0.0058 vs. 0.0061 eV). These improvements suggest that 3D geometric constraints provide valuable information for predicting frontier orbital energies, which exhibit high sensitivity to molecular conformation. For the dipole moment (μ) and polarizability (α) properties, the extended model demonstrates comparable but not superior performance relative to KANO, indicating that these properties may be more strongly influenced by electronic effects captured through alternative representations. Overall, the results confirm that 3D structural information serves as a valuable complementary source of information, particularly for geometry-dependent quantum properties such as frontier orbital energies.

Ablation study

To verify the effectiveness of each module of MolGramTreeNet, particularly the core role of molecular grammar tree representation, this study compared the performance of the complete model with models that had specific modules removed (MolGramTree, MolGCN, Transformer+GCN) on classification and regression tasks through ablation experiments. The experimental results are presented in Tables 5 and 6, with the specific analysis detailed later in discussion.

Table 2. Comparison results of MolGramTreeNet and baseline models in classification tasks

Task	Classification(ROC-AUC)						
Dataset	BACE	BBBP	Clintox	Tox21	ToxCast	SIDER	HIV
Molecule Numbers	1,513	2,039	1,478	7,831	8,575	1,427	41,127
GCN[2017]	0.854	0.877	0.807	0.772	0.650	0.638	0.744
MPNN[2017]	0.815	0.913	0.815	0.741	0.691	0.621	0.771
DMPNN[2020]	0.852	0.919	0.852	0.821	0.718	0.632	0.770
GraSeq[2020]	0.778	0.941	0.789	0.809	<u>0.734</u>	0.606	0.777
3D- informax[2022]	0.794	0.691	0.594	0.744	0.644	0.533	0.760
KCL[2022]	<u>0.924</u>	<u>0.956</u>	0.898	<u>0.821</u>	0.714	**0.671**	0.772
Uni-mol[2023]	0.857	0.729	0.919	0.791	0.696	<u>0.655</u>	**0.808**
KANO[2023]	**0.925**	**0.971**	<u>0.926</u>	0.819	0.718	0.651	0.765
MolCLR[2023]	0.850	0.724	0.880	0.784	0.691	0.597	0.778
GEM[2023]	0.856	0.724	0.901	0.781	0.692	0.672	<u>0.806</u>
MolGramTreeNet	0.878	<u>0.956</u>	**0.994**	**0.829**	**0.747**	0.620	0.775

Bold values indicate the best performance, and underlined values indicate the second-best performance.

a. MolGramTree: By ablating the GCN module, this variant relies exclusively on the CFG-parsed grammar tree encoded by the Transformer. It demonstrates superior performance to the topology-only MolGCN on five classification tasks and all regression tasks. Although it underperforms the full model by approximately 3.25%, its advantage over the graph-only baseline indicates that hierarchical semantic encoding is essential for prediction accuracy, even in the absence of 2D topological features.
b. MolGCN: Retaining only the 2D GCN encoder results in a significant performance decline, with an average decrease of 17.66% across 10 tasks compared to the full model. Relying solely on graph topology fails to capture the hierarchical semantics of functional groups explicitly encoded by the grammar tree. Consequently, the model struggles to represent complex relationships (e.g., solubility-related substructures), leading to inferior predictive performance.
c. Transformer + GCN: Directly feeding raw SMILES sequences into the Transformer without CFG parsing results in a 3.39% performance decline. The absence of grammatical constraints prevents the validation of cyclic structure validity, leading to errors in toxicity and free energy prediction (e.g., FreeSolv RMSE increased to 1.348). This demonstrates that sequence and graph modeling require the hierarchical regularization provided by the grammar tree to accurately predict property-dependent conformations.
d. MolGramTreeNet$_{att}$: We replaced channel concatenation with a cross-modal attention mechanism. Comparative analysis demonstrates that the simple concatenation strategy consistently outperforms attention fusion (average ROC-AUC improved by 2.3%, RMSE reduced by 13.9%). These results suggest that for molecular property prediction, the direct complementarity between grammar tree and graph features is best preserved through concatenation, whereas attention mechanisms introduce unnecessary complexity without yielding performance improvements.

Table 3. Comparison results of MolGramTreeNet and baseline models on regression tasks

Task	Regression(RMSE)		
Dataset	ESOL	FreeSolv	Lipophilicity
Molecule Numbers	1,128	642	4,200
GCN[2017]	1.431	2.900	0.712
MPNN[2017]	1.167	2.185	0.852
DMPNN[2020]	1.050	2.177	0.672
GraSeq[2020]	1.176	2.267	0.998
3D informax[2022]	0.798	1.855	0.880
KCL[2022]	<u>0.670</u>	<u>0.854</u>	0.789
Uni-mol[2023]	0.788	1.480	<u>0.603</u>
KANO[2023]	0.713	**0.521**	**0.489**
MolCLR[2023]	0.911	2.021	0.875
GEM[2023]	0.798	1.877	0.660
MolGramTreeNet	**0.547**	1.303	0.955

Bold values indicate the best performance, and underlined values indicate the second-best performance.

The ablation results confirm that the grammar tree serves as the core driver of performance. It explicitly encodes chemical hierarchies through CFG, addressing the absence of semantic constraints in GCNs and raw sequence models. The deep fusion of rule-based semantics 1D and spatial topology 2D synergistically enhances molecular representation. For example, the high receiver operating characteristic-area under the curve (ROC-AUC) score of 0.994 on ClinTox demonstrates the grammar tree's capability to identify valid cyclic structures, thereby preventing topological misclassifications.

Visualization

To intuitively demonstrate the learning efficiency and generalization capability of MolGramTreeNet on unseen data, this study employed t-SNE technology to visualize molecular embeddings from

Table 4. Comparison of the MolGramTreeNet+3D model on the QM9 dataset

Task	$\epsilon_{HOMO}(eV)$	$\epsilon_{LUMO}(eV)$	$\Delta\epsilon(eV)$	$\mu(D)$	$\alpha(bohr^3)$
MPNN[2017]	0.0435	0.0377	0.0698	**0.0302**	**0.0925**
3D Infomax[2021]	0.0430	0.0441	0.0592	<u>0.3502</u>	<u>0.3264</u>
KANO[2023]	<u>0.0044</u>	<u>0.0063</u>	<u>0.0061</u>	0.4179	0.5197
MolGramTreeNet+3D	**0.0042**	**0.0048**	**0.0058**	0.5104	0.5180

Bold values indicate the best performance, and underlined values indicate the second-best performance.

the ClinTox test set, as illustrated in Figure 4. The model used for generating these embeddings was trained exclusively on the ClinTox training and validation sets. A comparison of the embedding distributions before and after training reveals that the proposed MolGramTreeNet model exhibits strong category discrimination capability. For instance, the left panel of Figure 4 displays the distribution of the two labels—FDA approval status and toxicity—in the ClinTox dataset before training. Blue points represent FDA-approved molecules, pink points denote FDA-unapproved molecules, red points indicate toxic molecules, and green points signify non-toxic molecules. The molecular embeddings of these two labels exhibit substantial overlap. Specifically, blue FDA-approved molecules are dispersed among pink FDA-unapproved molecules, while green non-toxic molecules are intermixed with red toxic molecules, resulting in indistinct category boundaries. In contrast, the right panel of Figure 4 demonstrates distinct post-training clustering, where FDA-approved (blue) and non-toxic (green) molecules are clearly separated from toxic compounds (red). This separation confirms that MolGramTreeNet captures discriminative chemical semantics, effectively distinguishing between toxicity and approval status. These findings align with the model's outstanding performance, as evidenced by the ROC-AUC value of 0.994 in the ClinTox classification task (Table 2).

For comparison, we performed the same analysis on the second-best baseline, KANO (Figure S1). Although KANO achieves some separation, its clusters remain diffuse and overlapping. The sharper contrast in MolGramTreeNet's visualization demonstrates that our grammar tree-based approach learns significantly more discriminative features than leading baselines.

High-performance verification and in-depth experimental analysis of the ClinTox dataset

The molecular grammar generation rules designed in this study were not specifically optimized for ClinTox; instead, the same grammar generation rules as those for MoleculeNet in Table S1 were employed. However, the designed grammar tree constraint framework demonstrates strong suitability for the ClinTox dataset. As presented in Table 2, MolGramTreeNet achieved a breakthrough performance with a ROC-AUC of 0.994 in the ClinTox toxicity classification task, significantly outperforming the best baseline model, KANO (0.926) by 7.3%. This performance advantage originates from the explicit encoding of chemical rules and hierarchical structures by the grammar tree: for instance, the model accurately identifies toxic functional groups such as nitro ($-NO_2$) and thiocarbonyl (>C=S) through the CFG rule "bracket atom→element_symbols," thereby preventing misclassifications that may occur in traditional graph models due to their inability to capture the logic of functional group combinations.

To quantitatively illustrate this clustering, we tracked specific molecular trajectories in the normalized embedding space (Figure 4). Prior to training, a randomly selected FDA-approved molecule was positioned at (72.54, 2.16), intermixed with molecules from other categories. Following training, it migrated significantly to (8.94, 56.87), forming a tight cluster with other approved molecules. Similarly, for the CT_TOX label, a non-toxic molecule shifted from (59.15, 22.37) to (96.07, 14.22), transitioning from a mixed region to a distinct non-toxic cluster. These substantial coordinate transformations demonstrate MolGramTreeNet's ability to learn discriminative structural features, effectively segregating molecules based on toxicity and approval status.

Therefore, the t-SNE visualization empirically demonstrates that MolGramTreeNet optimizes molecular embeddings through multimodal fusion. The grammar tree module plays a crucial role in achieving this clear category separation, providing chemical semantic foundations that enhance both performance and interpretability.

Interpretability analysis: An attention-based case study

While the t-SNE visualization in the preceding section (Figure 4) demonstrates that MolGramTreeNet effectively separates

Table 5. Comparison of ROC-AUC in ablation experiments across seven classification tasks in MoleculeNet

Task	Classification(ROC-AUC)						
Dataset	BACE	BBBP	Clintox	Tox21	ToxCast	SIDER	HIV
Molecule Numbers	1,513	2,039	1,478	7,831	8,575	1,427	41,127
MolGramTreeNet	**0.878**	**0.956**	**0.994**	**0.829**	**0.747**	**0.620**	**0.775**
MolGramTree	<u>0.867</u>	<u>0.949</u>	0.987	0.778	0.706	<u>0.615</u>	0.727
MolGCN	0.738	0.871	0.848	<u>0.823</u>	0.652	0.603	<u>0.764</u>
Transformer+GCN	0.849	0.943	<u>0.991</u>	0.811	0.675	0.601	0.752
MolGramTreeNet$_{att}$	0.854	0.947	0.989	0.784	<u>0.741</u>	0.612	0.739

Bold values indicate the best performance, and underlined values indicate the second-best performance.

Table 6. Comparison of RMSE for ablation experiments on three regression tasks in MoleculeNet

Task	Regression(RMSE)		
Dataset	ESOL	FreeSolv	Lipophilicity
Molecule Numbers	1,128	642	4,200
MolGramTreeNet	**0.547**	**1.303**	**0.955**
MolGramTree	<u>0.549</u>	1.458	1.040
MolGCN	0.702	2.240	1.139
Transformer+GCN	0.584	<u>1.348</u>	<u>0.964</u>
$\text{MolGramTreeNet}_{att}$	0.551	1.576	1.130

Bold values indicate the best performance, and underlined values indicate the second-best performance.

molecules into distinct clusters based on their properties, this visualization alone is insufficient to explain how the model arrives at these decisions. To provide deeper, experimentally validated insights into the model's inner workings and to support our claim of providing chemical insights, we conducted an interpretability analysis by visualizing the internal attention mechanism of the Tree Sequence Encoder (Transformer module).

This analysis focuses on the attention weights of the special [GLO] token. This token is prepended to every rule sequence and is designed to aggregate information from all CFG rules, serving as the final, comprehensive vector representation of the entire molecule's grammar tree. This aggregated vector is what the model ultimately uses for prediction. By examining the attention scores the [GLO] token assigns to each CFG rule in the sequence, we can precisely quantify which grammatical and structural components the model deems most important for a given task. To validate our hypothesis that the model learns to identify specific, chemically meaningful toxicophores, we performed an expert-assessable case study using two representative molecules from the ClinTox dataset. We selected CCO (ethanol) as a simple, non-toxic control molecule. As our primary test case, we selected C1 = NC2 = C(N1)C(=S)N=CN2 (mercaptopurine).

This molecule was specifically chosen because it contains the well-known thiocarbonyl (>C=S) functional group. A thiocarbonyl is a chemical moiety in which a carbon atom is connected to a sulfur atom via a double bond (analogous to a >C=O carbonyl group). In the SMILES notation C1 = NC2 = C(N1)C(=S)N=CN2, this functional group is represented by the C(=S) fragment, where the carbon atom (part of the purine ring) is double-bonded to a sulfur atom in a branch. Our analysis tests whether the model can associate the CFG rules used to build this C(=S) fragment with toxicity.

The results presented in Figure 5 support this hypothesis. For non-toxic ethanol (Figure 5A), attention is diffusely distributed across generic structural rules, reflecting the absence of specific toxicity markers. In contrast, mercaptopurine (Figure 5B) exhibits a highly specific pattern where the model assigns peak attention

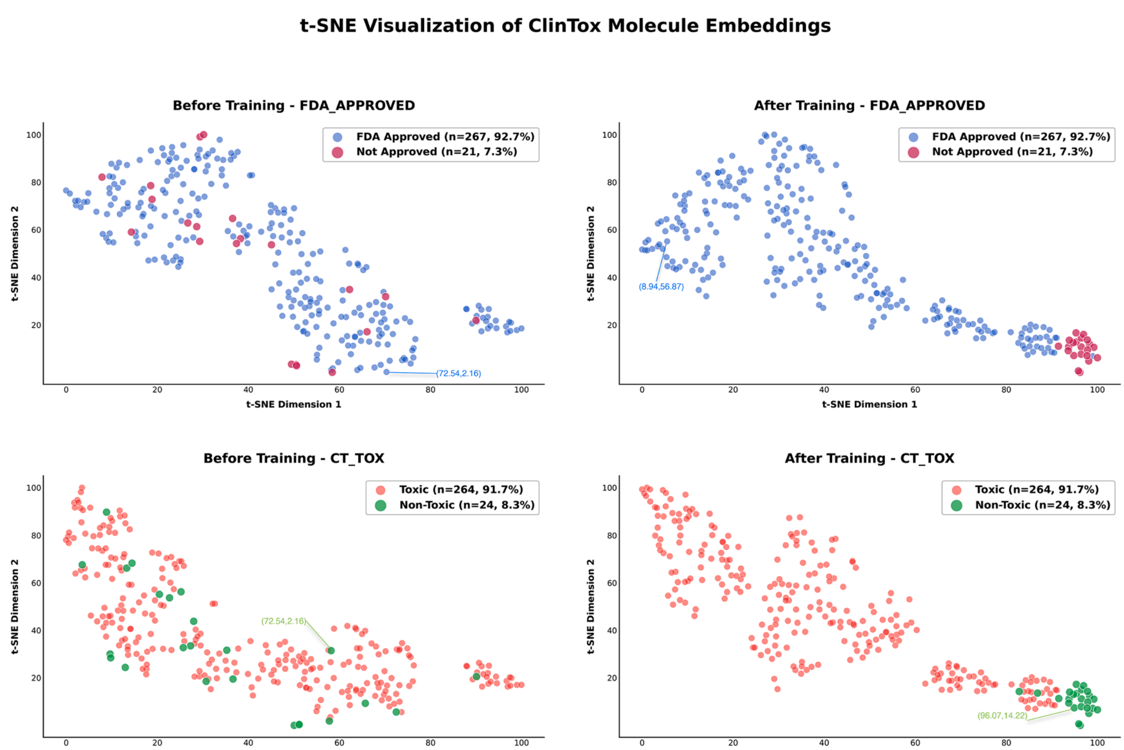

Figure 4. Visualization of the Clintox dataset for classification tasks (test set), illustrating how the model learns to distinguish between molecules of different categories

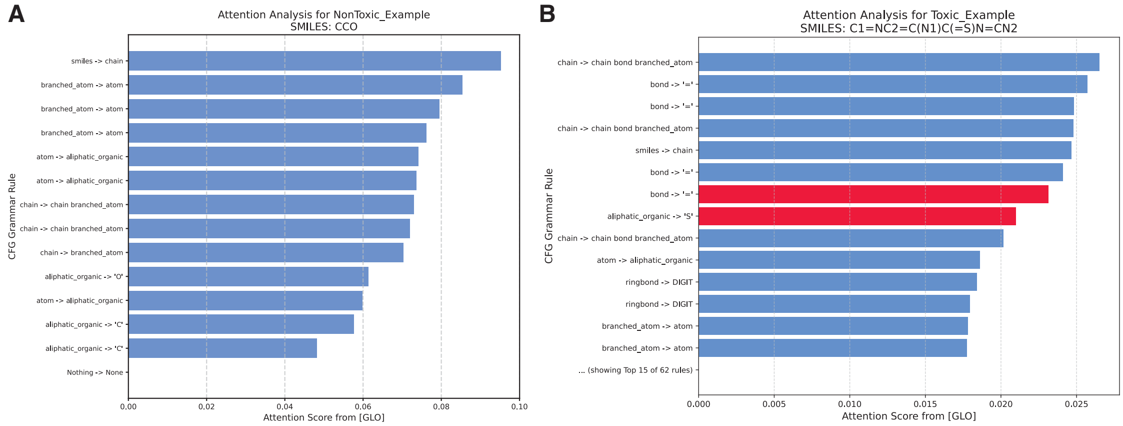

Figure 5. Attention analysis of MolGramTreeNet on ClinTox Samples

(A) Attention plot for the non-toxic molecule "CCO." All CFG rules are displayed.

(B) Truncated attention plot for the toxic molecule "C1 = NC2 = C(N1)C(=S)N=CN2," showing the top 15 most attended rules for clarity in the main text. The rules corresponding to the ">C=S" toxic functional group are highlighted in red. The complete attention plot for this molecule (displaying all 62 rules) is provided in Figure S2. Note: The x axis (attention score) scales for (A and B) differ intentionally (0.00–0.10 for (A), 0.00–0.025 for (B)) because molecular complexity influences the absolute attention distribution. Normalizing the scales for each plot independently represents the standard method for clearly visualizing the internal relative importance of rules for each specific molecule.

scores to the rules defining the thiocarbonyl group: bond → " = " and aliphatic_organic → "S." This demonstrates that MolGramTreeNet extends beyond opaque feature extraction, successfully learning to identify and prioritize chemically meaningful toxicophores through their grammatical construction.

Context-free grammar grammar rule perturbation experiment

To further investigate the core role of the CFG-based grammar tree representation and its contribution to model performance, we designed a set of CFG rule perturbation experiments. We aimed to verify whether the model's performance is directly dependent on the integrity of the chemical grammar information carried by the CFG sequence.

Experimental Design: We employed a "Random Replace" strategy. For the rule index sequence parsed from the CFG, we randomly selected a fraction of indices (0%, 10%, 30%, 50%, and 70%) and replaced them with other valid rule indices randomly sampled from the grammar dictionary. This experiment was conducted on the BBBP (classification) and ESOL (regression) benchmarks.

The results are presented in Figure 6. On the BBBP dataset (Figure 6A), the model's performance exhibited a clear dependence on the perturbation rate. As the replacement rate increased from 0% (baseline, ROC-AUC: 0.956) to 70% (ROC-AUC: 0.923), the model's classification accuracy displayed a distinct, monotonic downward trend. Similarly, on the ESOL dataset (Figure 6B), we observed a parallel trend. As the replacement rate increased, the model's prediction error (RMSE) significantly and consistently increased from 0.547 at 0% to 0.698 at 70%. This dose-effect relationship, observed consistently across both classification and regression tasks, strongly confirms that the CFG grammar tree is not a redundant feature. The decline in performance directly corresponds to the loss of grammatical-semantic information, demonstrating that MolGramTreeNet genuinely learns and relies on the precise chemical syntax and hierarchical structure provided by the CFG.

Furthermore, we note that while the performance degradation is significant, the model does not fail completely even at high perturbation rates. This highlights the robustness of our multimodal fusion framework. We attribute this to the dual-path architecture of MolGramTreeNet: when the information quality of the 1D Transformer path degrades due to the perturbed input, the 2D GCN path continues to extract accurate topological features from the original, unperturbed molecular graph. The stable representation from the GCN path compensates for the loss in the 1D path, further validating our hypothesis of multimodal feature complementarity.

Analysis of robustness to simplified molecular-input line-entry system variability

A key challenge for sequence-based models is the non-unique representation of molecules (e.g., "CCO" vs. "OCC"). We investigated how this input variability affects our CFG-Transformer encoder. We designed a "Canonicalized Model" as a control. In this model, all SMILES strings were first converted into their single, unique canonical representation via a deterministic algorithm before being fed to the CFG encoder. This process ensures a single, robust sequence input for every molecule, thereby eliminating all variability. We compared this model to our "Original Model" (which uses raw SMILES).

As shown in Table 7, the Original Model significantly outperforms the Canonicalized Model. This demonstrates that MolGramTreeNet utilizes SMILES variability as an effective implicit data augmentation technique. The invariant GCN path provides a topological anchor, compelling the variant Transformer path to learn a more generalized semantic representation by mapping multiple distinct input sequences to the same molecule, thereby mitigating overfitting.

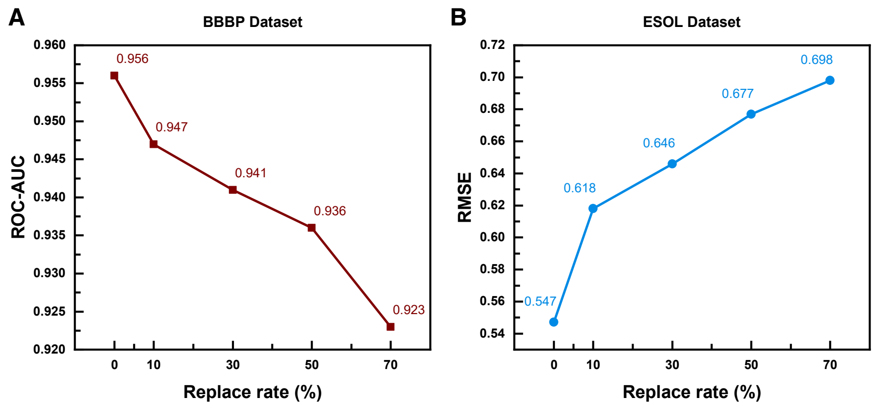

Figure 6. Effect of the CFG rule replacement rate on model performance

(A) ROC-AUC on the BBBP dataset.

(B) RMSE on the ESOL dataset. Data points represent the mean of 5 independent experimental runs.

Case study

To validate the practical utility of MolGramTreeNet in real-world pharmaceutical applications, we conducted a targeted case study focused on the safety profiling of potential hepatitis B virus (HBV) inhibitors. Hepatotoxicity (liver toxicity) remains a primary cause of drug attrition during antiviral therapeutic development. Consequently, deploying accurate in silico models to filter high-risk candidates early can significantly reduce experimental costs and streamline the drug discovery pipeline.

We constructed a specialized hepatotoxicity dataset containing 2,271 bioactive compounds extracted from the ChEMBL database.[40] These compounds were selected based on their available cytotoxicity data (CC50) against normal liver cells. To formulate this as a binary classification task, we adopted a rigorous toxicological threshold: compounds exhibiting a CC50 value below 30 μM were categorized as toxic (indicating high hepatotoxicity), whereas those with values of 30 μM or higher were classified as non-toxic. This threshold aligns with common cytotoxicity screening practices and is consistent with established computational toxicology studies, which have employed similar thresholds (e.g., approximately 40 μM) to differentiate toxic from non-toxic compounds.[41]

The model was trained on this curated dataset to assess its ability to discriminate between safe and toxic antiviral candidates. As detailed in Table 8, MolGramTreeNet demonstrated superior predictive capabilities compared to established baselines. Specifically, our model achieved an ROC-AUC of 0.824, surpassing the best-performing baseline model (GIN, 0.738) by a substantial margin of 8.6%. This significant improvement suggests that by explicitly encoding chemical grammar and hierarchical structures, MolGramTreeNet captures subtle structural motifs associated with drug-induced liver injury more effectively than topology-only approaches.

Table 7. Effect of SMILES variability on model performance

Dataset	Original Model	Canonicalized Model
BBBP	0.956	0.873
Clintox	0.994	0.922
ESOL	0.547	0.710

DISCUSSION

This study proposes a molecular representation and property prediction model, MolGramTreeNet, based on grammar tree constraints. In molecular property prediction research, the core innovation of MolGramTreeNet lies in its novel architecture, which achieves deep integration of chemical grammar rules with a multimodal feature fusion framework. Our unique contribution is the construction of this framework that innovatively combines grammar trees (via Transformer) and 2D graphs (via GCN). By using CFG to parse SMILES, the model explicitly encodes the hierarchical structure of molecules—such as the nested relationships between branches and rings. This approach fundamentally addresses a key limitation of traditional GNNs, which rely solely on topology while neglecting chemical hierarchies, and simultaneously ensures the chemical validity of the underlying representations.

The multimodal fusion serves as the central mechanism driving the model's performance. The one-dimensional (1D) grammar tree sequence, encoded by a Transformer, captures the hierarchical logic of molecular construction. Concurrently, the two-dimensional (2D) molecular graph, encoded by a GCN, extracts the topological features of atomic bonds. This synergistic integration of distinct feature types is critical. For instance, on the ESOL solubility prediction task, our full model (RMSE = 0.547) significantly outperforms the GCN-only variant

Table 8. Comparison of classification models for the HBV dataset

Dataset	CC_{50}
Molecule Numbers	2,271
Transformer	0.695
MPNN	0.716
GIN	0.738
MolGramTreeNet	**0.824**
MolGCN	0.803
MolGramTree	<u>0.821</u>
Transformer+GCN	0.814

Bold values indicate the best performance, and underlined values indicate the second-best performance.

(MolGCN, RMSE = 0.702). This result provides clear evidence that the model captures the synergistic influence of functional group hierarchy (from the tree) and atomic bonds (from the graph) on solubility. It validates our hypothesis that the complementary integration of 1D and 2D features is essential for predicting complex chemical properties, thereby ensuring a comprehensive molecular representation.

The experimental results indicate that, compared to eight baseline models such as GCN and message-passing neural network (MPNN), MolGramTreeNet improves average performance by 2.86%–21.47% across ten tasks. On the ESOL dataset, performance improves by 18.36% compared to the suboptimal model KCL, while on the ClinTox dataset, performance improves by 7.34% compared to the suboptimal model KANO. Additionally, a visualization experiment was conducted on the ClinTox dataset to further validate the model's interpretability. The results clearly demonstrate that MolGramTreeNet exhibits strong category discrimination capability. Furthermore, a case study was performed to evaluate hepatotoxicity (CC50) on the HBV dataset, showing an 8.6% improvement in performance compared to the best baseline model. These results confirm the effectiveness of integrating molecular grammar tree representation with a multi-modal framework for molecular property prediction tasks.

Limitations of the study

Despite the promising performance of MolGramTreeNet, this study has several limitations that present opportunities for future research.

1. The current grammar tree representation is constructed based on a context-free grammar (CFG) applied to one-dimensional (1D) SMILES sequences. This study did not explore the potential of constructing analogous grammar trees directly from the 2D molecular graphs.
2. Our model was trained end-to-end without a pre-training phase. We did not investigate the potential benefits of incorporating our grammar tree-guided representation into a self-supervised or pre-training framework. Such an approach could enhance the model's generalization capabilities, particularly in low-data scenarios.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Li Wang (wangli@ntu.edu.cn).

Materials availability

This study did not generate any new unique reagents.

Data and code availability

- The data analyzed in this study were obtained from publicly accessible sources, and all data sources have been detailed in the "key resources table" within the article.
- The original code for this study has been deposited on GitHub (https://github.com/NTU-MedAI/MolGramTreeNet). Relevant digital object identifiers (DOIs) are also provided in the "key resources table" in the article.
- Additional information required to reanalyze the data reported in this article is available from the corresponding author upon request.

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AUTHOR CONTRIBUTIONS

Y.Z.: methodology, software, validation, and writing – original draft. W.L.: investigation, resources, data curation, and writing – original draft. H.Z.: investigation and writing – original draft. J.S.: validation and formal analysis. S.P.: visualization and software. M.Y.: data curation and resources. K.N.: conceptualization, writing – review and editing, supervision, and funding acquisition. L.W.: conceptualization, writing – review & editing, supervision, and funding acquisition.

DECLARATION OF INTERESTS

Compliance with Ethics Requirements This article does not contain any studies with human or animal subjects. The authors declare no competing financial interest.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS
- METHOD DETAILS
  - Construction and encoding of molecular grammar tree
  - Context-free grammar
  - Molecular grammar
  - Syntax tree structure to numerical sequence
  - A transformer-based one-dimensional syntax tree digital sequence feature encoder
  - GCN-based 2D feature encoder for molecular graphs
  - Multidimensional feature fusion encoding
  - 3D feature extension and architecture
  - Loss function
  - Datasets
  - Baseline models
- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
MoleculeNet	An online database of MoleculeNet	https://moleculenet.org/
QM9	An online database of QM9	https://quantum-machine.org/datasets/
HBV	Gaulton et al.[40]	https://www.ebi.ac.uk/chembl/
Software and algorithms		
MolGramTreeNet	This Paper	https://github.com/NTU-MedAI/MolGramTreeNet
CFG	Kusner et al.[37]	https://arxiv.org/abs/1703.01925
Transformer Architecture	Vaswani et al.[5]	https://arxiv.org/abs/1706.03762
GCN Architecture	Kipf et al.[23]	https://arxiv.org/abs/1805.11973
3D Molecular Encoder	Zhou et al.[42]	https://openreview.net/forum?id=6K2RM6wVqKu
Python (version 3.8.0)	Python Software Foundation	https://www.python.org/
PyTorch (version 2.0.1)	Open-source software	https://pytorch.org/
RDKit (version 2022.03.3)	Open-source software	http://www.rdkit.org/
Other		
NVIDIA GeForce RTX 2080 Ti	NVIDIA Corporation	https://www.nvidia.com/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Omitted as our study is strictly computational and does not involve the use of live biological models, human subjects, or primary cell lines. All data used in this study were obtained from previously published, anonymized public databases (MoleculeNet, QM9, and ChEMBL), as detailed in the key resources table.

METHOD DETAILS

Construction and encoding of molecular grammar tree

In this section, the study presents the methodological background for converting context-free grammar (CFG)-based molecular grammar trees into digital sequences. This includes an introduction to context-free grammar, an overview of grammar-based molecular tree representation for SMILES, and a description of the process for extracting rule indices from grammar trees into digital sequences. These components will be systematically introduced in the following subsections.

Context-free grammar

Formal grammars have long served as the foundation for various language modeling tasks, including semantic interpretation of natural language, dialogue understanding, and machine translation. They are primarily based on the principle that groups of words belong to the same constituent unit, and different constituent units can be hierarchically combined to convey specific meanings.

This study posits that molecular representation based on SMILES strings constitutes a formal grammar system, where each character functions analogously to a word and each string represents a sentence. Consequently, in molecular representation tasks, this study employs CFG to parse the grammatical structure of molecular SMILES strings. The underlying concept of CFG is that sentences can be recursively constructed from smaller phrases, and these constituents can be organized according to a semantic hierarchy. Formally, a CFG is defined by the quadruple $G = (V, \Sigma, R, S)$, where: V denotes the set of non-terminal symbols, representing decomposable units in the grammatical structure; Σ represents the set of terminal symbols, corresponding to the basic atomic elements of the language; R constitutes the set of production rules, which describe the decomposition logic of non-terminal symbols; and S is the start symbol, representing the highest-level structure of the language.Each production rule follows the form $A \rightarrow \alpha$, where $\alpha \in V$ is a non-terminal symbol, and α is a sequence of terminal or non-terminal symbols.

The CFG used for molecular SMILES representation includes characters in SMILES strings as terminal symbols and meta-information about molecular structures as non-terminal symbols. In Table 1, we present a representative subset of the SMILES grammar used in the work of this study. The symbols defining the CFG in this example are as follows.

- *V*: chain, branched_atom, bond, atom, ringbond, BB, RB, branch, digit, aromatic_organic, aliphatic_organic
- *Σ*: { = , C, N, O, 1}
- *R*: production rules 1 through 20 in Table
- *S*: SMILES

Molecular grammar

To concretely illustrate this CFG parsing process, we validate the grammar tree representation method using a molecular SMILES example from the BBBP dataset: divinyl ether, with the string representation "C=COC = C". This molecule was selected for its structural simplicity, which facilitates a clearer demonstration of the grammar tree construction. The corresponding parse tree structure is illustrated in Figure 2. The grammar tree representation was derived from the production rules presented in Table 1 (note: these production rules were specifically simplified for "C=COC = C"; complete and standardized grammar production rules are provided in Table S1).

The construction of the grammar tree for "C=COC = C" follows a bottom-up, left-corner parsing strategy. Beginning with the leftmost terminal symbol "C", it is mapped to the non-terminal aliphatic_organic using Rule 16 (aliphatic_organic → C). Subsequently, Rule 15 (atom → aliphatic_organic) generates an atom node, followed by Rule 6 (branched_atom → atom) to form branched_atom_1. Finally, Rule 4 (chain → branched_atom) creates the minimal molecular chain chain_1. The adjacent double bond " = " is processed using Rule 13 (bond → =) to form bond_1. The next "C" undergoes a similar derivation to form branched_atom_2, after which chain_1, bond_1, and branched_atom_2 are combined using Rule 3 (chain → chain bond branched_atom) to form chain_2, representing the fragment "C=C". The terminal "O" is mapped to aliphatic_organic via Rule 17 (aliphatic_organic → O), then to atom via Rule 15, and to branched_atom_3 via Rule 6. chain_2 and branched_atom_3 are merged using Rule 2 (chain → chain branched_atom) to form chain_3. The subsequent "C" is processed similarly to form branched_atom_4, and chain_3 and branched_atom_4 are combined via Rule 2 to form chain_4, representing "C=COC". Finally, the last " = " and "C" are processed to form bond_2 and branched_atom_5, respectively. chain_4, bond_2, and branched_atom_5 are combined using Rule 3 to form the complete chain chain_5, which is then mapped to the root node SMILES using Rule 1 (SMILES → chain). This step-by-step construction ensures that every derivation adheres to the CFG rules. thereby establishing a hierarchical and grammatically valid representation of the molecule.

This CFG-based parsing method systematically elucidates the grammatical logic of SMILES strings, mapping hierarchical features of molecular structures—including atomic connection order, chemical bond types, and cyclic or branched structures—into computable grammar tree representations. This structured parsing paradigm not only reveals explicit associations between molecular composition rules but also, through the dynamic combination capability of grammar tree nodes, provides an interpretable theoretical framework for automated modeling of molecular representations. Consequently, it significantly enhances the model's capacity to capture and generalize chemical semantics, thereby facilitating subsequent molecular property prediction tasks.

Syntax tree structure to numerical sequence

Following the construction of the grammar tree, the subsequent step involves its transformation into a computable digital sequence. The core of converting a molecular grammar tree into a digital sequence involves transforming the hierarchical information of the tree structure into a computable discrete sequence through the indexing of context-free grammar (CFG) rules. The grammar tree construction employs a bottom-up strategy, while the encoding into a digital sequence follows a top-down order of rule application, ensuring the sequence accurately reflects the hierarchical decomposition logic of the tree structure.

The encoding method for the molecular grammar tree in Figure 2 utilizes indexed mapping of grammar rules. First, each CFG rule is assigned a unique digital index to create a rule-index mapping table. For example, consider the simplified grammar in Table 1.

- Rule 1: SMILES → chain → index 1
- Rule 6: branched_atom → atom → index 6
- Rule 15: atom → aliphatic_organic → index 15
- Rule 17: aliphatic_organic → O → index 17

Each decomposition rule of non-terminal symbols corresponds to a unique integer, forming a "grammar dictionary" of molecular structures.

After constructing the grammar tree, the tree structure is traversed from top to bottom and left to right, with indices extracted in the order of rule application, as illustrated in Figure 3 for "C=COC = C". Specifically, the process begins with the root node SMILES, where Rule 1 (SMILES → chain) is applied, and index 1 is recorded. The node chain_5 is decomposed into three child nodes (chain_4, bond_2, branched_atom_5) using Rule 3, and index 3 is recorded. Subsequently, traversing the leftmost node, chain_4 is decomposed into chain_3 and branched_atom_4 via Rule 2, with index 2 recorded. This process continues recursively: chain_3 is decomposed into chain_2 and branched_atom_3 using Rule 2 (index 2), and chain_2 is decomposed into chain_1, bond_1, and branched_atom_2 via Rule 3 (index 3). For chain_1, it is decomposed into branched_atom_1 using Rule 4 (index 4), followed by the decomposition of branched_atom_1 into atom via Rule 6 (index 6), atom into aliphatic_organic via Rule 15 (index 15), and aliphatic_organic into "C" via Rule 16 (index 16). Similarly, the remaining nodes are traversed, and their corresponding rule indices are recorded. The resulting digital sequence for “C=COC = C” is [1, 3, 2, 2, 3, 4, 6, 15, 16, 13, 6, 15, 16, 6, 15, 17, 6, 15, 16, 13, 6, 15, 16]. This sequence fully captures the hierarchical structure and rule applications of the grammar tree.

A transformer-based one-dimensional syntax tree digital sequence feature encoder

To capture the hierarchical semantics and contextual dependencies of the digital sequence of molecular grammar trees, this study employs a multi-layer Transformer encoder as the sequence feature extraction model, as illustrated in Figure 1B.

This module is based on the self-attention mechanism of the Transformer, which adaptively learns long-range dependencies between rule indices. The architecture primarily consists of an embedding layer (Embedding_Layer), multi-layer Transformer encoder layers, and a global feature extraction module. First, the digital sequence x obtained after index rule extraction enters the initial network layer of this module: the embedding layer, which comprises token embedding and positional embedding. Token embedding maps rule indices (integers) to d_{model}-dimensional semantic vectors to learn the fundamental representation of rules, while positional embedding encodes sequence position information to capture the order dependence of rule application. The two embeddings are combined and output as initial features through layer normalization:

$$H_0 = LayerNorm\left(E_{tok}(x) + E_{pos}(p)\right) \quad \text{(Equation 1)}$$

where x is the rule index sequence, p is the position index, and $H_0 \in \mathbb{R}^{N \times d_{model}}$ (N is the sequence length, d_{model} is the model dimension).

The input then passes through multiple Transformer encoder layers. The attention mechanism serves as the core component of the Transformer architecture, enabling the capture of dynamic correlations between sequence elements. By simulating the cognitive process of selectively attending to relevant information, the model adaptively identifies semantic dependencies across different positions. In the context of molecular grammar tree encoding, Multi-Head Self-Attention enhances this capability: for the hidden state $H_{l\text{-}1}$, the model first generates query (Q), key (K), and value (V) matrices through linear transformation, which can be expressed as:

$$Q = H_{l-1}W^Q \quad \text{(Equation 2)}$$

$$K = H_{l-1}W^K \quad \text{(Equation 3)}$$

$$V = H_{l-1}W^V \quad \text{(Equation 4)}$$

where W^Q, W^K, W^V are learnable weights. For the input hidden feature H, it can be split into h low-dimensional subspaces, with each head dimension $d_k = d_{model}/h$. In each subspace, the correlation weight is calculated through scaled dot-product attention, that is, the query is multiplied by the transpose of the key, divided by $\sqrt{d_k}$ to alleviate gradient vanishing, normalized by the Softmax function and multiplied by the value matrix to obtain the single-head output:

$$head_i = Softmax\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right)\boldsymbol{V_i}(i = 1, \ldots, h) \quad \text{(Equation 5)}$$

Subsequently, the outputs of all heads are concatenated and the dimension is restored through linear transformation W^{out}, that is:

$$MultiHead(H_{l-1}) = Concat(head_1, \ldots, head_h)W^{out} \quad \text{(Equation 6)}$$

This captures local and global correlations between molecular rules from multiple perspectives, such as atomic connections and ring hierarchical structures.

After completing the attention calculation, the feedforward network layer performs a nonlinear transformation on each positional feature to enhance the model’s expressive capability:

$$FFN(H_l') = ReLU(H_l' W_l^1)W_l^2 \quad \text{(Equation 7)}$$

Output through residual connection and layer normalization:

$$H_l = LayerNorm(H_{l-1} + FFN(LayerNorm(H_{l-1} + MultiHead(H_{l-1})))) \quad \text{(Equation 8)}$$

Finally, in the global feature extraction stage, the hidden state at the first position of the sequence is extracted as the global tree sequence feature:

$$h_{tree} = H_N[:, 0, :] \quad \text{(Equation 9)}$$

It is integrated with molecular graph features (GCN module) for downstream prediction.

In adapting to molecular grammar trees, the embedding layer learns the grammatical roles of rules, and the self-attention mechanism captures the hierarchical dependencies among rules. The model employs an end-to-end training approach, wherein parameters are randomly initialized without pre-training, with a focus on chemical semantic modeling. For the regularized representation of molecular grammar trees, the discrete rule index sequence is first mapped to continuous high-dimensional feature vectors, and the feature distribution is stabilized through normalization operations. Subsequently, the multi-head self-attention mechanism dynamically captures semantic correlations between rules from parallel perspectives, followed by the application of a masking mechanism to filter out invalid information in the sequence padding positions. The encoder layer then connects the self-attention module and the feedforward network in series, utilizing residual connections and layer normalization to construct a deep feature extraction unit that aggregates local and global semantics layer by layer. After iterative optimization by multi-layer encoders, the global feature of the molecular grammar tree with dimension d_{model} is output via global pooling, as denoted by h_{tree} in Equation 9. This feature is summed and fused with the two-dimensional structural features from the molecular graph encoder, enhancing the model's ability to jointly model molecular hierarchical semantics and topological structures through the complementarity of multi-modal features.

The advantages of this encoder include: (1) employing Transformer self-attention to effectively capture long-range hierarchical dependencies between rules, demonstrating superior performance compared to traditional models such as LSTM; (2) featuring a lightweight and interpretable architecture, where self-attention weights visualize rule importance to facilitate molecular grammar logic analysis; and (3) generating tree sequence features that complement 2D molecular graph features, thereby enhancing chemical semantic modeling and improving prediction performance. The Transformer_Encoder enables efficient encoding of molecular grammar tree sequences, provides one-dimensional feature representations for multimodal fusion in MolGramTreeNet, strengthens the interpretation and representation of hierarchical molecular structure semantics, and supports downstream molecular property prediction tasks.

GCN-based 2D feature encoder for molecular graphs

To extract the two-dimensional structural features of molecules, this study employs a Graph Convolutional Network (GCN) as the molecular graph encoder, as illustrated in Figure 1C. A molecule is modeled as a graph $G=(V,E)$, where nodes V represent atoms. The node features consist of two components: atomic types (such as carbon, nitrogen, oxygen, etc.), totaling 119 types including a mask token, and chirality tags, totaling three types describing the chiral properties of atoms. Edges E represent chemical bonds, with features comprising two components: bond types and bond directions. There are five bond types (including single bonds, double bonds, aromatic bonds, and self-loops), and three bond directions primarily used to describe the directional information of bonds. Through multilayer graph convolution operations, the GCN aggregates structural information of nodes and their neighborhoods layer by layer, ultimately generating a molecular representation vector containing global structural features. Its core layer iteratively updates node representations as:

$$h_v^{(l+1)} = \sigma\left(\sum_{u \in \mathcal{N}(v) \cup v} \frac{1}{c_{v,u}} \cdot \left(W^{(l)} \cdot h_u^{(l)} + E_{e_{u,v}}\right)\right) \quad \text{(Equation 10)}$$

where $h_v^{(l)}$ is the feature vector of node v at layer l, $\mathcal{N}(v)$ is the neighbor set of node v, $c_{v,u} = \sqrt{|\mathcal{N}(v)| \cdot |\mathcal{N}(u)|}$, $W^{(l)}$ is a learnable weight matrix, $\boldsymbol{E}_{e_{u,v}}$ is the embedding representation of edge $e_{u,v}$, and σ is a non-linear activation function.

For the input molecular graph, feature extraction is initially performed to obtain x, edge_index, edge_attr, and batch, which describe the structure and features of the molecular graph. These features are subsequently input into the node embedding layer, which maps discrete atomic features (type, chirality) to continuous embedding vectors. Given the input atomic feature vector $x = [x_{type}, x_{chirality}]$, where $x_{type} \in Z^1$ (atomic type index) and $x_{chirality} \in Z^1$ (chirality tag index), atomic feature embedding is performed as follows:

$$x_v = E_{type}(t_v) + E_{chirality}(c_v) \quad \text{(Equation 11)}$$

where $t_v \in \{1, \ldots, 119\}$ is the atomic type, $c_v \in \{1,2,3\}$ is the chirality tag, and E_{type} and $E_{chirality}$ are learnable embedding matrices. Subsequently, bond feature embedding is performed:

$$e_{u,v} = E_{bond\_type}(b_{u,v}) + E_{direction}(d_{u,v}) \quad (12)$$

where $b_{u,v} \in \{1, \ldots, 5\}$ is the bond type, $d_{u,v} \in \{1,2,3\}$ is the bond direction, and E_{bond_type} and $E_{chirality}$ are learnable embedding matrices.

The GCN encoder takes the aforementioned Equations 11 and 12 as inputs and aggregates neighborhood information layer by layer through stacking L graph convolution units, thereby enabling iterative updates of atomic features and the construction of global representations. In each layer, the message-passing mechanism first aggregates the feature contributions of neighboring nodes u for node v: the l-th layer feature $h_u^{(l)}$ of node u undergoes linear transformation via the layer-specific weight matrix $W_1^{(l)}$, is combined with the edge attribute embedding $e_{u,v}$, and is then multiplied by the normalization coefficient $c_{v,u}$ derived from the degree matrix to form a message:

$$m_v^l = \sum_{u \in \mathcal{N}(v) \cup v} \frac{1}{c_{v,u}} \cdot \left(W^{(l)} \cdot h_u^{(l)} + E_{e_{u,v}}\right) \quad \text{(Equation 13)}$$

Subsequently, nodes process the message through an activation function σ (e.g., ReLU) and batch normalization to obtain the next layer's node features:

$$h_v^{(l+1)} = BatchNorm\left(\sigma\left(m_v^{(l)}\right)\right) \quad \text{(Equation 14)}$$

This enhances training stability and introduces nonlinearity. After L-layer iteration, global feature aggregation employs mean pooling to summarize the final features $\{h_v^{(L)}\}_{v \in V}$ of all nodes into a graph-level representation:

$$h_{\text{graph}} = \frac{1}{|\mathcal{V}|}\sum\nolimits_{v \in V} h_v^{(L)} \quad \text{(Equation 15)}$$

Subsequently, via linear transformation, the global graph feature is ultimately projected into a unified feature space:

$$h_{gcn} = W_{out} . h_{graph} + b_{out} \quad (16)$$

where b_{out} is the bias vector of the linear transformation, which constitutes a learnable parameter of the model. The output global feature representation of the molecular graph, as shown in Equation 16 (i.e., h_{gcn}), provides the two-dimensional (2D) structural feature representation of the molecule for MolGramTreeNet. This representation complements the one-dimensional (1D) sequence feature h_{tree} derived from the grammar tree, and their fusion collectively enhances the model's predictive performance for molecular properties.

Multidimensional feature fusion encoding

To achieve multi-dimensional semantic modeling of molecules, MolGramTreeNet integrates the feature representations from the Transformer_Encoder-based 1D grammar tree sequence encoder and the GCN-based 2D molecular graph encoder through a channel concatenation strategy. Specifically, the global grammar tree feature output by the Transformer_Encoder is $h_{\text{tree}} = H_N[:,0,:]$ with a dimension of d_{model}, capturing the hierarchical grammatical logic of molecular structures. The molecular graph feature output by the GCN is $h_{gcn} = W_{out}.h_{graph}+b_{out}$ with a dimension of d_{feat}, representing the 2D structural dependencies between atoms. These two features are concatenated along the channel dimension to form a unified multi-modal feature vector:

$$h_{fusion} = Concat\left(h_{tree} + h_{gcn}\right) \quad \text{(Equation 17)}$$

The output dimension is $d_{model} + d_{\text{feat}}$.

Compared with complex attention fusion or weight learning methods, channel concatenation offers the advantages of simplicity and efficiency. This approach does not require additional parameter tuning, directly preserves the complete information from both modalities, and maintains low computational complexity, making it suitable for end-to-end training. The fused features can be directly fed into downstream task modules, such as classifiers or regression layers, and mapped to the target dimension through fully connected layers to enable efficient prediction of molecular properties. In summary, as a core component of MolGramTreeNet, the multi-dimensional feature fusion module effectively combines the 1D grammatical logic and 2D structural information of molecules through a simple yet efficient channel concatenation strategy, forming a comprehensive semantic representation. This integration provides a robust feature foundation for downstream molecular property prediction tasks and significantly enhances the model's predictive performance and generalization capability.

3D feature extension and architecture

In the extended study, we integrated three-dimensional structural information into the original MolGramTreeNet framework, constructing MolGramTreeNet+3D. This model incorporates a computationally efficient three-dimensional encoder designed to process atomic coordinates and elemental information. The three-dimensional module accepts atomic numbers and three-dimensional coordinates as input. Atomic types are first embedded into a continuous vector space, then concatenated with spatial coordinates. The combined features are processed through fully connected layers incorporating rectified linear unit activation functions and dropout regularization:

$$NodeFeatures = FC(Concat[Embed(Z), r]) \quad \text{(Equation 18)}$$

Where Zrepresents atomic numbers and r denotes 3D coordinates. Global mean pooling aggregates node-level features into molecular representations.

The 3D features are concatenated with existing 1D grammar tree and 2D graph features:

$$h_{fusion} = Concat\left(h_{tree}, h_{treegraph}, h_{3d}\right) \quad \text{(Equation 19)}$$

This comprehensive representation captures syntactic, topological, and geometric molecular characteristics.

Loss function

To construct the prediction model, MolGramTreeNet employs a multi-channel dual-dimensional drug feature encoder capable of fusing molecular grammar tree information with two distinct dimensions. The downstream tasks consist of a regression task and a classification task. Cross-entropy loss and RMSE loss functions are utilized to optimize MolGramTreeNet. Cross-entropy loss is applied to classification tasks and has been evaluated on all classification tasks in MoleculeNet. For regression tasks, this study employs the RMSE loss function, which has been tested on the ESOL, FreeSolv, and Lipophilicity datasets. Additionally, for the 3D-extended experiments on the QM9 dataset, the Mean Absolute Error (MAE) loss function is employed to align with established benchmarks.

The total loss functions of MolGramTreeNet are:

$$Loss_{CE} = -\sum N\,log\widehat{N} - \lambda(1 - N)log(1 - \widehat{N}) \qquad \text{(Equation 20)}$$

$$Loss_{RMSE} = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(\widehat{y_i} - y_i)^2} \qquad \text{(Equation 21)}$$

$$Loss_{MAE} = \frac{1}{K}\sum_{i=1}^{n}|y_k - \widehat{y_k}| \qquad \text{(Equation 22)}$$

where N, M, K is the true label of the sample, $\widehat{N}$ is the predicted probability of the sample by the model, y is the class label of the sample, and $\widehat{y_i}$, $\widehat{y_k}$ is the predicted value of the sample.

Datasets

We comprehensively evaluated MolGramTreeNet using ten standard benchmarks from the MoleculeNet suite, covering diverse molecular domains including quantum mechanics, physical chemistry, and biophysics. These benchmarks comprised seven classification tasks and three regression challenges.

For the classification tasks, we focused on bioactivity and toxicity. Specifically, we employed BACE (1,513 entries) to identify beta-secretase 1 inhibitors and BBBP (2,039 entries) to assess blood-brain barrier penetration. Viral inhibition capabilities were evaluated using the HIV dataset (41,127 compounds), while potential adverse drug reactions were analyzed via the SIDER database (1,427 entries). To rigorously test toxicity prediction, we utilized three specialized datasets: ClinTox (1,478 compounds), which distinguishes between FDA-approved drugs and those failing clinical trials due to toxicity; Tox21 (7,831 compounds), derived from the 2014 Data Challenge; and ToxCast (8,575 compounds), consisting of data from advanced high-throughput screening technologies.

For the regression tasks, we aimed to predict key physicochemical properties. Water solubility and hydration free energy were evaluated using the ESOL (1,128 molecules) and FreeSolv (642 molecules) datasets, respectively. Additionally, the Lipophilicity dataset (4,200 molecules) was included to assess the model's ability to predict octanol-water partition coefficients.

To ensure robust evaluation and facilitate feature learning under data-constrained conditions, all datasets were partitioned into training, validation, and testing subsets following an 8:1:1 ratio. To mitigate potential biases from random splitting, we performed independent experiments using ten distinct random seeds (ranging from 2016 to 2025) and reported the averaged performance metrics.

Additionally, to evaluate the impact of three-dimensional structural information, we employed the QM9 dataset, a widely used benchmark for predicting molecular quantum mechanical properties. The QM9 dataset contains approximately 134,000 stable organic molecules with up to nine heavy atoms (carbon, oxygen, nitrogen, fluorine), providing a comprehensive testbed for quantum chemical systems. Following standard practice, we partitioned the dataset into a training set (110,000 molecules), a validation set (10,000 molecules), and a test set (10,831 molecules). All energy-related properties are reported in electron volts (eV).

For hepatotoxicity screening, we utilized the HBV dataset derived from ChEMBL,[40] a leading open-access bioactivity repository managed by the European Bioinformatics Institute (EMBL-EBI). This comprehensive database aggregates over 2.4 million drug-like compounds, primarily extracted from peer-reviewed scientific literature to ensure high data diversity and validity. Its reliability is further supported by meticulous manual curation and extensive interoperability with external chemical registries, such as PubChem and ChemSpider, which enables rigorous cross-validation of bioactive data.

Baseline models

Supervised learning (SL) serves as the foundation of computational drug discovery, particularly for mapping molecular structures to their corresponding physicochemical or biological properties. These data-driven methods excel in classification and regression tasks when supported by reliable, high-quality labeled datasets. To rigorously evaluate the performance of our proposed framework, we benchmarked MolGramTreeNet against a diverse set of ten established algorithms.

GCN (Graph Convolutional Network): This model extends the concept of convolution to non-Euclidean graph data. It operates by iteratively aggregating feature vectors from adjacent nodes to update the representation of the central atom. A global pooling (readout) operation is subsequently applied to synthesize these node-level embeddings into a comprehensive molecular vector representation.[43]

MPNN (Message Passing Neural Network): Distinguished from standard node-centric GNNs, MPNN explicitly incorporates edge attributes (chemical bonds) into the message-passing phase. By processing information from both nodes and their connecting edges during the update step, it captures a more comprehensive set of 2D topological dependencies within the molecule.[44]

DMPNN (Dynamic Message Passing Neural Network): This architecture addresses the issue of signal redundancy (tottering) in message passing. By constructing a directed graph where messages are propagated along directed bonds rather than atoms, DMPNN learns distinct vector representations for atoms and bonds, thereby improving the encoding of the molecular structure.[16]

GraSeq: As a representative multimodal baseline, GraSeq employs a dual-encoder strategy. It utilizes a graph convolutional network (GCN) to extract topological features from the molecular graph and a bidirectional long short-term memory (Bi-LSTM) network to capture sequential dependencies from the SMILES string. These two distinct feature sets are then concatenated for the final property prediction.[31]

KCL (Knowledge-Centric Learning): This framework improves molecular representation by incorporating external domain knowledge. It constructs a knowledge graph (KG) based on molecular structures and employs a pre-training strategy that assigns weights to atomic nodes according to their relevance within the KG, thereby integrating chemical priors into the model.[45]

3D Infomax: Recognizing the limitations of 2D topology, this method employs a contrastive pre-training approach. It aims to maximize the mutual information between the latent representations of a 2D molecular graph and its corresponding 3D geometric conformer, effectively transferring 3D spatial insights into the 2D encoder.[46]

Uni-mol: This model represents a shift toward direct three-dimensional processing. Utilizing a Transformer-based backbone, Uni-mol takes three-dimensional atomic coordinates as input and is pre-trained on tasks such as denoising molecular conformations, enabling it to capture precise geometric information directly rather than inferring it from a graph.[42]

KANO: This approach leverages an element-centric knowledge graph (ElementKG) to guide the pre-training process. By utilizing element-specific masking strategies and functional prompts during fine-tuning, KANO enhances the capture of intrinsic chemical semantics and elemental relationships within molecules.[47]

MolCLR: Based on the graph contrastive learning paradigm, MolCLR learns robust representations by maximizing the agreement between different views of the same molecule. These views are generated through specific graph augmentation techniques, such as atom masking, bond deletion, and subgraph removal.[18]

GEM (Geometry-aware Message Passing Network): this method integrates geometric information through specialized self-supervised learning tasks. The network is trained to predict geometric parameters, including bond lengths and angles, thereby explicitly embedding three-dimensional spatial constraints into the message-passing framework.[48]

QUANTIFICATION AND STATISTICAL ANALYSIS

All experiments were conducted in a Linux environment, primarily using Python 3.8 for programming. The deep learning framework employed was PyTorch 2.0.1 (cu117), and molecular processing and feature extraction were performed with RDKit 2022.03.3. Computations were accelerated using an NVIDIA GeForce RTX 2080 Ti GPU (11 GB VRAM). To ensure the robustness of the results, all datasets underwent repeated experiments with 10 different random seeds (ranging from 2016 to 2025). Each dataset was randomly split into training, validation, and test sets in an 8:1:1 ratio. Detailed experimental configurations and dataset information can be found in the "Experimental Parameters" subsection of the supplemental information and in the key resources table. Model performance was evaluated using the following quantitative metrics. For classification tasks, the primary metric was the Area Under the Receiver Operating Characteristic Curve, with Binary Cross-Entropy Loss used as the training loss function. For regression tasks, the Root-Mean-Square Error served as the main performance measure; training utilized Mean Squared Error Loss, which was then converted to RMSE for final evaluation.

To quantify the performance improvement of MolGramTreeNet over the baseline models, this study applied a unified calculation method. The specific calculation formula is provided in the "Performance Calculation Methodology" subsection of the supplemental information.